Generation & Validation of a Mechanical Bidirectional Sweep Signal to Bolster Human T-Cell Proliferation for Cancer Immunotherapy

Pola Urmanski[1,2], Andreas Kaimis[1,2], Mohammad Kader[1,2], Matthew Kuan[1,2], Anusree Chakraborty[1,2], James Han[1,3], Clinton Rubin[2], M. Ete Chan[1,2]

[1]*Bioengineering Education, Application and Research (BEAR) Vertically Integrated Project Team*
[2]*Department of Biomedical Engineering,* Stony Brook University
[3]*Department of Biology,* Stony Brook University

Introduction

Motivation:

- During exercise, the body experiences a wide range of vibrations that influence different physiological processes.
- Low-Intensity Vibration (LIV) delivers mechanical oscillations in the range of ~10-150Hz to biological systems.
  - Key parameters include: displacement, acceleration, duration, refractory period, and number of treatments.
- Previously studied effects of 0.7g, 30 Hz vibrations (2×1hr/day for 5 days) on T-cell proliferation to enhance biomanufacturing yields for CAR-T therapies.

Objective:

- To develop a Frequency and Amplitude Modulated Sweep **(FAMS)** signal that applies a band of several frequencies, providing a more physiologically relevant model of the vibrations experienced during exercise.

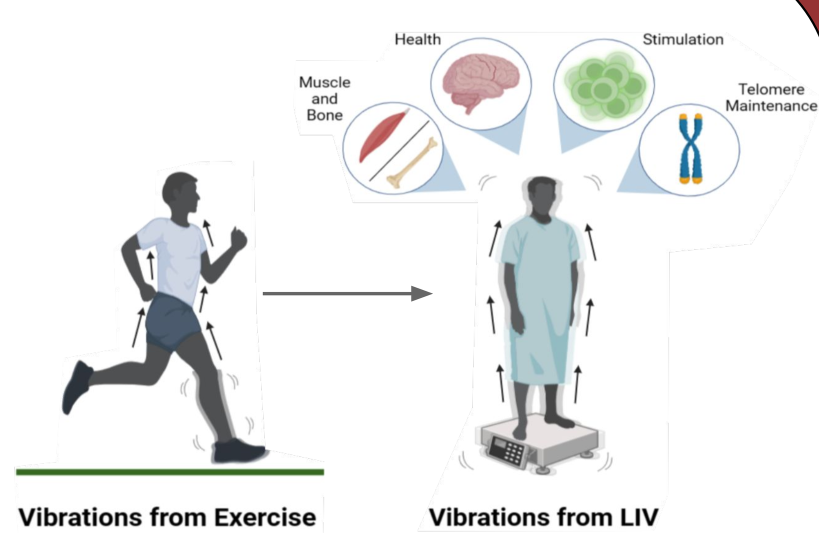

Fig 1. Various LIV applications.

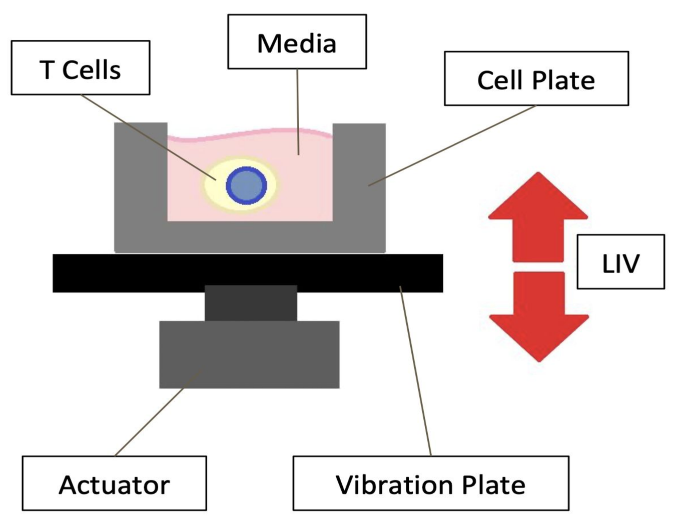

Fig 2. LIV experimental setup for stimulating suspension cell.

Data Analysis

Signal Validation

Features:

- Acceleration measured using *enDAQ Slamstick,* piezoelectric accelerometer.
- Python script features several libraries: enDAQ, plotly, pandas, numpy, scipy.
- Python Script Quality Control:
  - Acceleration & Displacement Visualization
  - Peak Acceleration Detection
  - Individual Chirp Analysis
  - Ramp Time, Rate of Change, and Frequency Verification
  - Noise detection

Results:

- Visualized quality control figures and reports using plotly.

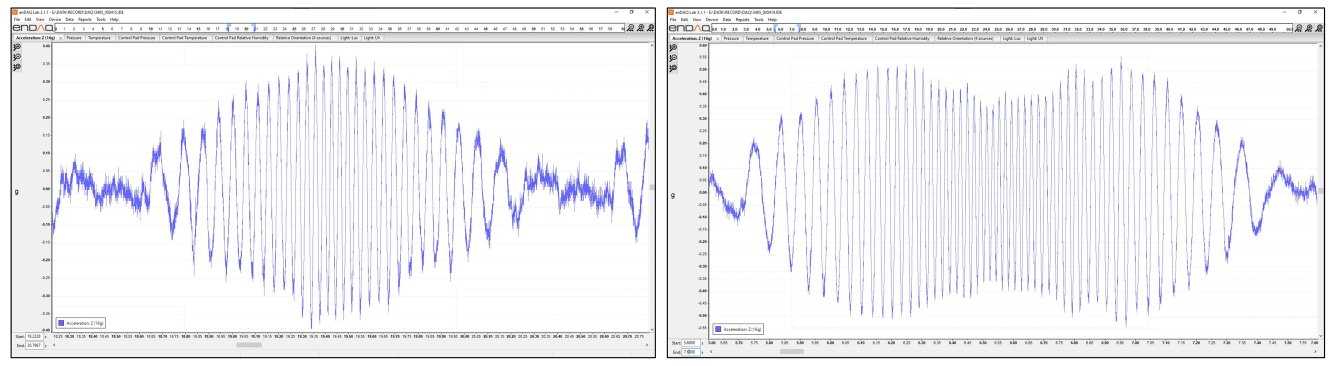

Fig 4. enDAQ software showing a successful sweep from 0–30 Hz (left) and an unexpected dip in acceleration at higher frequencies, peaking at 50 Hz (right).

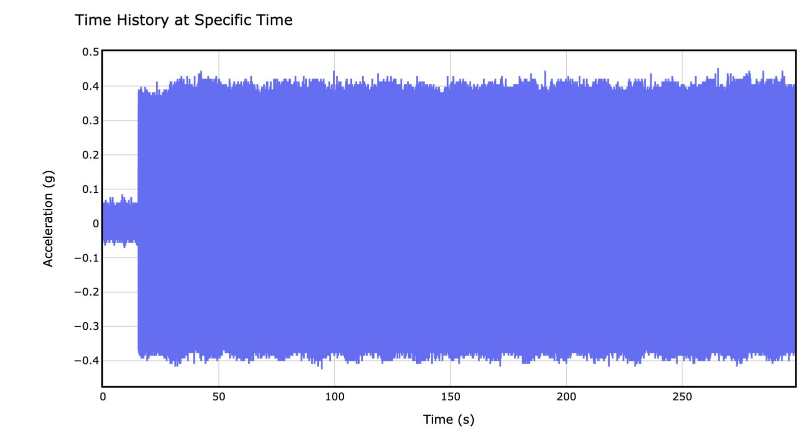

Fig 5. Signal validation using a Python script to display a test, static 30 Hz signal at 0.7g p-p.

Methods & Results

LIV Device & FAMS Signal Generation

System Features:

- LabVIEW software
- Microcontroller—Arduino UNO
- Cylindrical Linear Voice Coil Actuator (solenoid mechanism)
- Triple-Axis Accelerometer
- Rubber Bands (to secure culture flasks and plates)

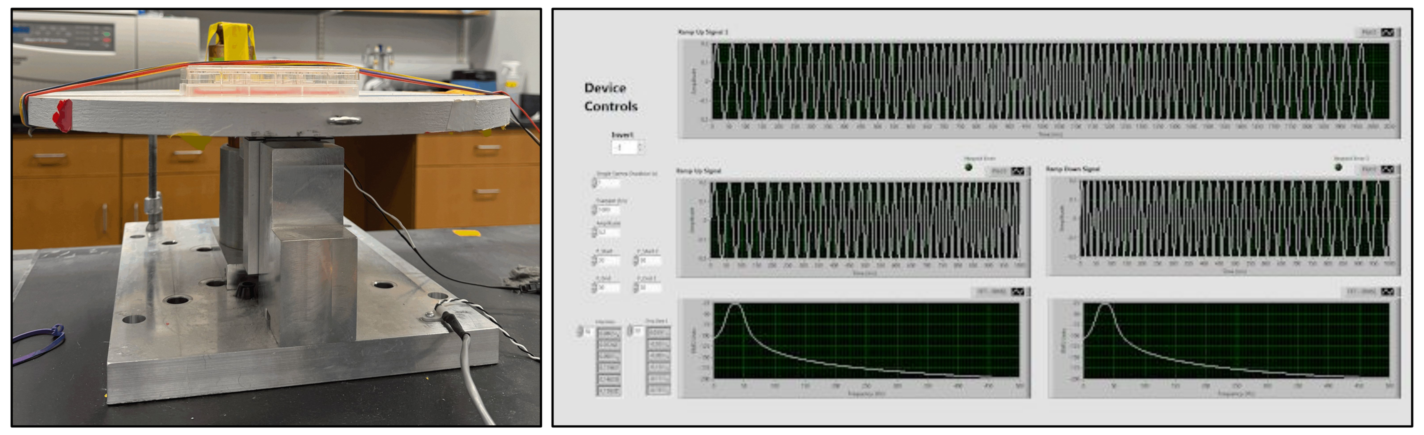

Fig 3. Tabletop device setup (left) and LabVIEW virtual interface (right).

Results:

- A bilateral sweep signal is generated via a LabVIEW VI, then converted, amplified, and sent to the actuator to produce vibrations at the desired frequencies and accelerations.
- The Labview VI allows custom sweep signal inputs for selecting frequencies, amplitude, timing, and sampling rate.
- The device can successfully produce the expected vibration profile from 0 to 35 Hz.
- However, oscillating artifacts are seen at frequencies of 36 Hz and higher.

Conclusion & Future Research

- The current LIV cell stimulation device successfully performs a bilateral sweep from 0 to 35 Hz; however, further development is needed to achieve consistent vibration delivery in the 36–50 Hz range.
- The Python-based quality control script requires continued development to ensure consistent delivery of sweep frequencies and accelerations, and to verify that vibration intensity does not exceed 0.9g (peak-to-peak).
- Integrating live acceleration feedback with a calibrated accelerometer will streamline testing by removing the need to export Slamstick data to a computer.
- **Upcoming in vitro study**: T-cells will be stimulated with the FAMS signal (20–50 Hz) in two 1-hour sessions per day over 5 days, mirroring the schedule of foundational studies.

References

- https://bpb-us-e1.wpmucdn.com/you.stonybrook.edu/dist/c/4606/files/2020/11/BMES-2020-Presentation.pdf
- Low intensity mechanical signals promote proliferation in a cell-specific manner: Tailoring a non-drug strategy to enhance biomanufacturing yields. M. Ete Chan, Lia Strait, Christopher Ashdown, Sishir Pasumarthy, Abdullah Hassan, Steven Crimarco, Chanpreet Singh, Vihitaben S Patel, Gabriel Pagnotti, Omor Khan, Gunes Uzer, Clinton T Rubin bioRxiv 2023.07.05.547864; doi: https://doi.org/10.1101/2023.07.05.547864

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